Program 7: Deploy a web application to kubernetes using Minikube

S-1: In the terminal
- mkdir prog7
- cd prog7
- code .

S-2 In VS code, create a nginx-deployment.yaml

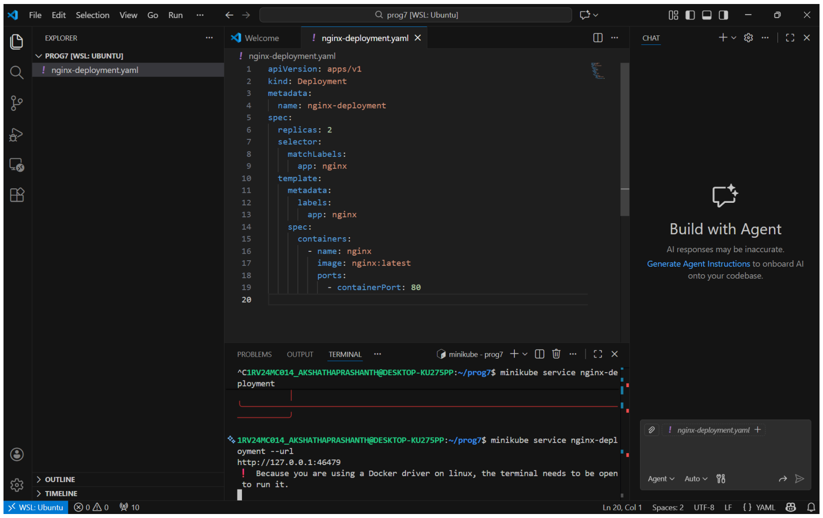
```yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: nginx-deployment
spec:
  replicas: 2
  selector:
    matchLabels:
      app: nginx
  template:
    metadata:
      labels:
        app: nginx
    spec:
      containers:
        - name: nginx
          image: nginx:latest
          ports:
            - containerPort: 80
```

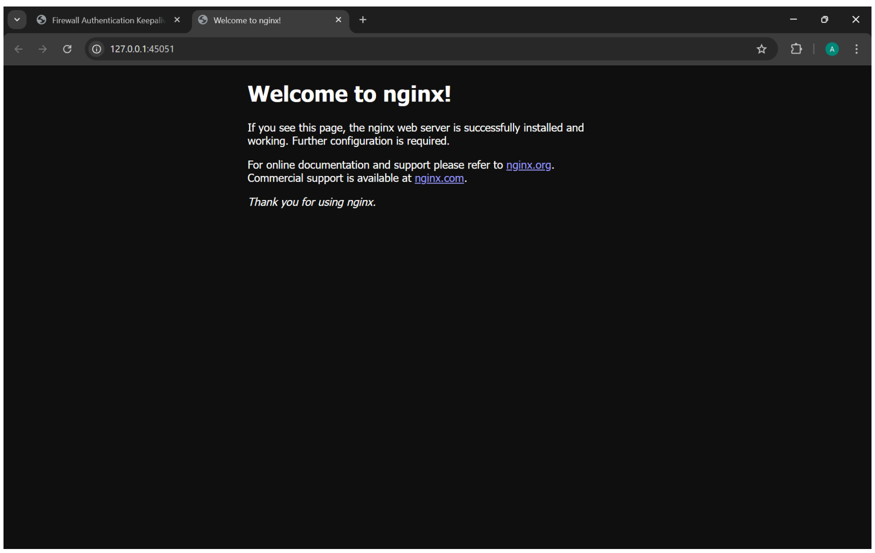
S-3: Start the minikube and check the status
Minikube start
minikube status

S-4: Apply the nginx-deployment.yaml
kubectl apply -f nginx-deployment.yaml

S-5: check status
Kubectl get deployments
kubectl get pods

S-6: Expose the deployment as a service
kubectl expose deployment nginx-deployment pods/nginx 3000:3000


File Edit Selection View Go Run
prog7 [WSL: Ubuntu]
EXPLORER
PROG7 [WSL: UBUNTU]
nginx-deployment.yaml
Welcome
nginx-deployment.yaml
apiVersion: apps/v1
kind: Deployment
metadata:
name: nginx-deployment
spec:
replicas: 2
selector:
matchLabels:
app: nginx
template:
metadata:
labels:
app: nginx
spec:
containers:
- name: nginx
image: nginx:latest
ports:
- containerPort: 80
CHAT
Build with Agent
AI responses may be inaccurate.
Generate Agent Instructions to onboard AI onto your codebase.
PROBLEMS OUTPUT TERMINAL
minikube - prog7
^C1RV24MC014_AKSHATHAPRASHANTH@DESKTOP-KU275PP:~/prog7$ minikube service nginx-deployment
1RV24MC014_AKSHATHAPRASHANTH@DESKTOP-KU275PP:~/prog7$ minikube service nginx-deployment --url
http://127.0.0.1:46479
❗ Because you are using a Docker driver on linux, the terminal needs to be open to run it.
Agent Auto
OUTLINE
TIMELINE
WSL: Ubuntu
Ln 20, Col 1 Spaces: 2 UTF-8 LF { } YAML


Firewall Authentication Keepalive
Welcome to nginx!
127.0.0.1:45051
Welcome to nginx!
If you see this page, the nginx web server is successfully installed and working. Further configuration is required.
For online documentation and support please refer to nginx.org.
Commercial support is available at nginx.com.
Thank you for using nginx.

Program 8: Create the POD YAML using node js as a base image

S-1: In the terminal

- mkdir prog8
- cd prog8
- code .

S-2 In VS code terminal, write a nodejs app -> server.js

```
const http= require("http");
const port= 3000;

const server= http.createServer((req,res)=>{
    res.end("Hello World! This is my first Node.js application running on Kubernetes");
});

server.listen(port, ()=>{
    console.log(`Server is running on port ${port}`);
});
```

S-3: Create a configmap.yaml

```
apiVersion: v1
kind: ConfigMap
metadata:
  name: nodejs-app-config
data:
  server.js: |
    const http = require("http");
    const port = 3000;

    const server = http.createServer((req, res) => {
      res.end("Hello from Node.js running inside Kubernetes!");
    });

    server.listen(port, () => {
      console.log(`Server running at port ${port}`);
    });
```

S-4: Create the POD YAML using node js as a base image

```yaml
apiVersion: v1
kind: Pod
metadata:
  name: program-8
  labels:
    app: nodejs
spec:
  containers:
    - name: nodejs
      image: node:18
      command: ["node", "/usr/src/app/server.js"]
      volumeMounts:
        - name: app-code
          mountPath: /usr/src/app
      ports:
        - containerPort: 3000
  volumes:
    - name: app-code
      configMap:
        name: nodejs-app-config
```

S-5: Apply the YAML files

kubectl apply -y nodejs-configmap.yaml
kubectl apply -y nodejs-pod.yaml

S-6: Check pod status

kubectl get pods

S-7: Access the nodejs app

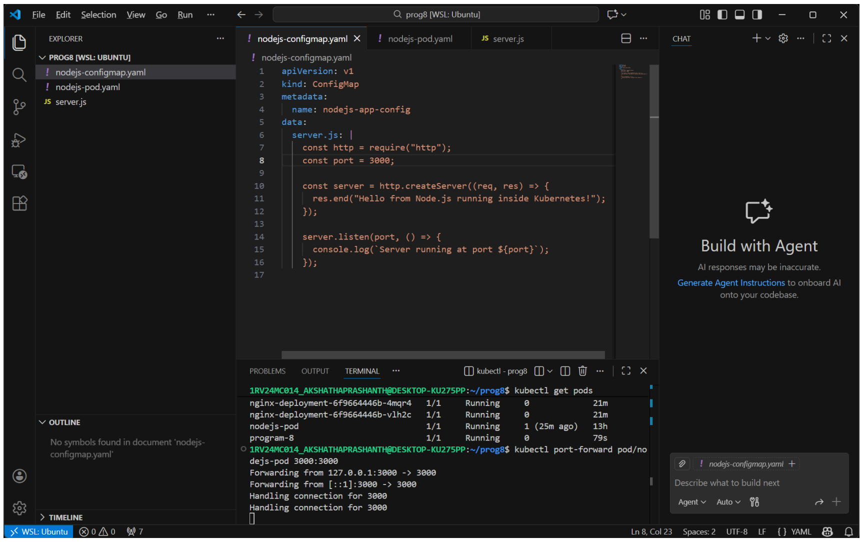
kubectl port-forward pod/nodejs -pod 3000:3000

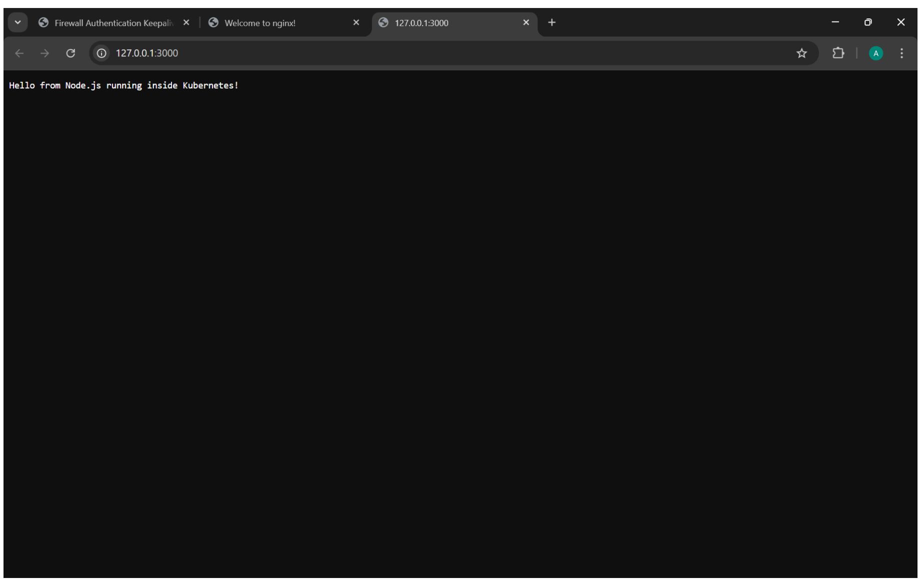
S-8 : Check the output in

http://localhost:3000

S-9: Delete all pods, services and deployments

kubectl delete pods –all
kubectl delete svc –all
kubectl delete deployments –all


prog8 [WSL: Ubuntu]
EXPLORER
PROG8 [WSL: UBUNTU]
nodejs-configmap.yaml
nodejs-pod.yaml
server.js
OUTLINE
No symbols found in document 'nodejs-configmap.yaml'
TIMELINE
nodejs-configmap.yaml
nodejs-pod.yaml
server.js
apiVersion: v1
kind: ConfigMap
metadata:
name: nodejs-app-config
data:
server.js: |
const http = require("http");
const port = 3000;
const server = http.createServer((req, res) => {
res.end("Hello from Node.js running inside Kubernetes!");
});
server.listen(port, () => {
console.log(`Server running at port ${port}`);
});
PROBLEMS
OUTPUT
TERMINAL
kubectl - prog8
1RV24MC014_AKSHATHAPRASHANTH@DESKTOP-KU275PP:~/prog8$ kubectl get pods
nginx-deployment-6f9664446b-4mqr4 1/1 Running 0 21m
nginx-deployment-6f9664446b-vlh2c 1/1 Running 0 21m
nodejs-pod 1/1 Running 1 (25m ago) 13h
program-8 1/1 Running 0 79s
1RV24MC014_AKSHATHAPRASHANTH@DESKTOP-KU275PP:~/prog8$ kubectl port-forward pod/nodejs-pod 3000:3000
Forwarding from 127.0.0.1:3000 -> 3000
Forwarding from [::1]:3000 -> 3000
Handling connection for 3000
Handling connection for 3000
CHAT
Build with Agent
AI responses may be inaccurate.
Generate Agent Instructions to onboard AI onto your codebase.
Describe what to build next
Agent
Auto
WSL: Ubuntu
Ln 8, Col 23
Spaces: 2
UTF-8
LF
YAML


Firewall Authentication Keepali
Welcome to nginx!
127.0.0.1:3000
127.0.0.1:3000
Hello from Node.js running inside Kubernetes!